

experience—to ensure these factors do not drive our estimates (e.g., adopters being younger and naturally on upward earnings trajectories). Second, we leverage our panel data to implement a difference-in-differences approach indexed to November 2022, the release date of ChatGPT. This allows us to control for time-invariant differences between workers (e.g., high-ability adopters who would have earned more regardless) and examine whether adopters experienced differential changes after the launch of ChatGPT. Crucially, the event-study design allows us to assess whether adopters were on distinct labor market trends before AI chatbots became available. Finally, we verify that our conclusions about the effects of worker-level use are robust when we rely only on the variation in chatbot use that stems from employer policies. Employer initiatives may be more plausibly unrelated to individual worker trajectories, as they reflect firm-level choices rather than worker self-selection into chatbot use.

A third challenge is timing: When should we date the arrival of AI chatbots, and when should we expect their effects on labor market outcomes to appear? We use the launch of ChatGPT in November 2022 as a before-and-after moment in an event-study design, marking a sharp rise in public awareness of AI chatbots.¹¹ We also separately examine effects for early adopters, identified as workers already using chatbots in our 2023 survey round, to allow more time for effects to materialize.¹² Even with rapid adoption, however, labor market effects may take time to materialize. We examine these adjustment dynamics in two ways. First, we implement a dynamic difference-in-differences design to trace how effects unfold monthly following the introduction of AI chatbots. Second, we analyze heterogeneity across occupations that vary in the flexibility of labor rules.

A final concern is that difference-in-differences identifies only *differential* effects for

¹¹This timing is motivated by the rapid uptake of ChatGPT among workers: most adopters began using it within a year of its launch (Humlum and Vestergaard, 2025). Appendix E.2 shows that ChatGPT remains the dominant AI chatbot to date. Moreover, event studies centered on its launch in November 2022 have become common in the literature (e.g., Brynjolfsson, Chandar and Chen (2025); Eisfeldt et al. (2024); Lichtinger and Hosseini Maasoum (2025); Schubert (2025); Teutloff et al. (2025)), facilitating comparison across studies.

¹²Our survey does not record the exact timing of adoption, precluding event studies indexed to workers' month of first use. However, even if such information were available, adoption is unlikely to represent a sharp before-and-after event, as workers typically ramp up usage gradually. This is another reason why we expect difference-in-differences effects to emerge over time rather than immediately.